

Machine Learning using Python

Module I: Basics

1. Introduction to machine learning
 2. different forms of learning;
 3. Basics of probability theory
 4. linear algebra and optimization
 5. Overview of target function representations
 6. Types of ML techniques
 7. hypothesis selection through cross validation.
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Module II: Regression Analysis

1. Linear regression
 2. ridge regression
 3. Lasso
 4. Bayesian regression,
 5. regression with basis functions
 6. Correlation, covariance, Mahalanobis distance, Mahalanobis distance, Minkowski distance, distance metric,
 7. Jaccard coefficient, missing values, feature transformations,
 8. Geometrical interpretation of Euclidean,
 9. Mahalanobis distance, dealing with uncertainty.
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Module III: Statistical Estimation

1. Maximum likelihood estimation (MLE) theory and example using binomial distribution, Maximum likelihood estimation (MLE) of univariate Gaussian,
 2. generative vs discriminative models,
 3. Maximum likelihood estimation of bivariate Gaussian distribution,
 4. sufficient statistics, Bayesian Learning
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Module IV: Computational Learning Theory

1. Computational Learning Theory: PAC Learning,
 2. VC Dimension,
 3. Bias/Variance Tradeoff.
 4. Classification Methods: Linear Discriminant Analysis,
 5. Logistic regression,
 6. Perceptrons,
 7. Large margin classification,
 8. Kernel methods,
 9. Support Vector Machines.
 10. Classification and Regression Trees,
 11. Multi-layer Perceptrons and Back propagation
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Module V: Clustering Techniques

1. Computational Learning Theory: PAC Learning,
2. VC Dimension,
3. Bias/Variance Tradeoff.

4. Classification Methods: Linear Discriminant Analysis,
5. Logistic regression, Perceptrons
6. Large margin classification,
7. Kernel methods,
8. Support Vector Machines.
9. Classification and Regression Trees
10. Multi-layer Perceptrons and Back propagation Clustering: Partitional Clustering - k-means, k-medoids;
11. Hierarchical Clustering - Agglomerative, Divisive, Distance measures;
12. Density based clustering - DBScan;
13. Spectral clustering. Frequent Pattern Mining: Apriori Algorithm; FP-Growth